

nRF5340 Production Programming

Application Note



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SEMICONDUCTOR

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Revision history

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January 2025	Editorial updates
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January 2021	First release

1 Introduction

This document provides information on writing software to the nRF5340 devices and is intended for developers of flash programming tools.

It serves as a starting point for the nRF5340 device support in production tools and accelerates the engineering process of supporting the nRF5340 devices. This document describes a robust method for programming devices. You might not need to follow every step in some cases, for example, if the device has never been programmed before and its flash is completely erased, or if the device is unprotected.

2 Programming flow

The diagram shows the flow of production programming under normal circumstances.

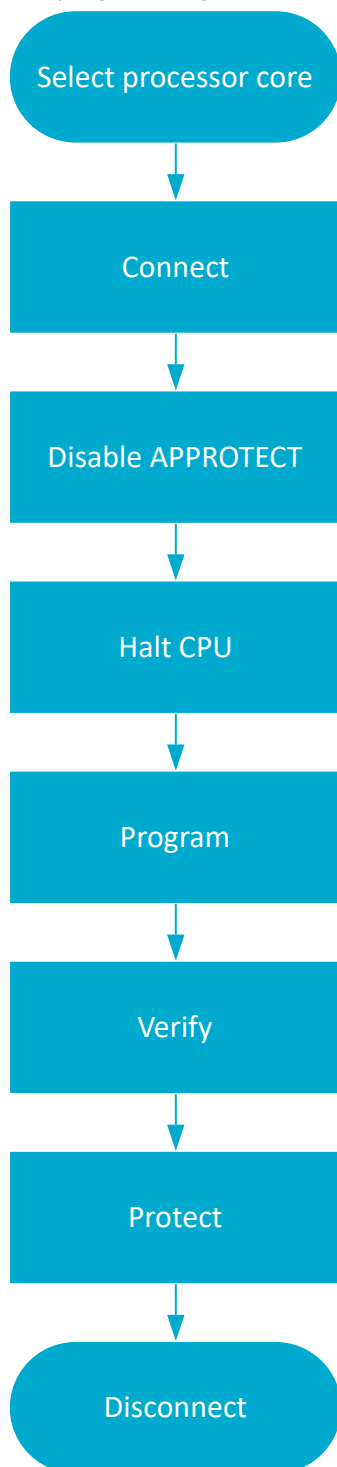


Figure 1: Normal programming flow

3 Selecting the processor core

The nRF5340 has two processors cores, the application core and the network core.

When communicating with the device using the *Serial Wire Debug Port (SW-DP)*, the application core at AHB-AP=0 is selected by default. The network core can be found at AHB-AP=1. To select the target processor, set AHB-AP to the corresponding value.

4 Disabling APPROTECT

If the device has access port protection enabled and is not erase protected, you can disable access port protection with a *Control Access Port (CTRL-AP)* erase all operation.

1. Write 1 to CTRLAP.ERASEALL.
2. Wait until the value of CTRLAP.ERASEALLSTATUS becomes 1.

Note: nRF5340 devices have two CTRL-APs, one for each core.

5 Halting the CPU

Use the standard *Serial Wire Debug (SWD)* Arm® CoreSight™ *Debug Access Port (DAP)* protocol to issue a halt command to the chip.

An application running on the device that was previously programmed can use the *Watchdog timer (WDT)*. The default configuration of the WDT will pause it, if the CPU is halted.

6 Connecting

Use the standard *SWD* Arm CoreSight *DAP* protocol to enter [debug interface mode](#).

Before the external debugger can access the CPU, it must first request the device to power up and make sure that the appropriate power domains are powered up. This is managed using the built-in `CxxxPWRUPREQ` and `CxxxPWRUPACK` features found in the DAP. If the debugger requests the debug domain or the complete system to power up, the device stays in debug interface mode.

Before connecting to the network core, check if it is in Force-OFF mode and not held in reset. To check if the network core is powered up, do an AHB read of AHB-AP 0 by targeting `RESET.NETWORK.FORCEOFF` (`0x50005614`) in the application core. If the readout is 0, the network core is not in Force-OFF mode. If the readout is 1, write 0 to it to exit Force-OFF mode.

7 Programming

When writing is enabled, the non-volatile memory is written by writing a word to a word-aligned address in the code or *User Information Configuration Registers (UICR)*. Only word-aligned writes are allowed. Byte or half-word-aligned writes result in a hard fault.

7.1 Writing data - SECUREAPPROTECT disabled

Use the standard *SWD* Arm CoreSight *DAP* protocol to write data into flash.

The *Non-volatile Memory Controller (NVMC)* peripheral has two different base addresses:

- Application core – 0x50039000
 - Network core – 0x41080000
1. Set the CONFIG register of the NVMC to `WEN.Wen` by writing 0x00000001 to the following addresses:
 - Application core - 0x50039504
 - Network core - 41080504

This enables writing to the non-volatile memory.

2. Read the READY register (0x50039400) of the NVMC until the value is 0x00000001. Read the following addresses:
 - Application core - 0x50039400
 - Network core - 0x41080400

When this value is read, the NVMC is ready and not currently performing any operations.

3. Write the data to the desired, word-aligned address.
4. Read the READY register of the NVMC until the value is 0x00000001 before continuing to ensure the write operation has completed. Read the following addresses:
 - Application core - 0x50039400
 - Network core - 0x41080400
5. Continue writing and then reading the READY register as necessary.
6. Set the CONFIG register of the NVMC to `WEN.Ren` by writing 0x00000000 to the following addresses:
 - Application core - 0x50039504
 - Network core - 41080504

This configures the non-volatile memory as read-only.

The ranges of writeable addresses are:

- *UICR* addresses (located in addresses 0x00FF8000 through 0xFF8FFC for the application core, 0x01FF8000 through 0x01FF8FFC for the network core)
- All program flash (located in addresses 0x00000000 through ((INFO.CODESIZE * INFO.CODEPAGESIZE) – 0x4))

You can write to flash using different methods. For nRF5340, a good flash algorithm should take around 11 seconds to write the entire flash to the application core, and around 3 seconds for the network core. If you cannot achieve this time, contact Nordic Semiconductor for assistance.

7.2 Verifying flash content

To verify the contents of flash after programming, use the standard SWD Arm CoreSight DAP protocol to read every address written and compare to expected values.

7.3 Enabling device protection

There are several ways to protect nRF5340 devices. *Access Port Protection (APPROTECT)* secures the access port, *Erase Protection (ERASEPROTECT)* stops the device from being erased, and *Secure Access Port Protection (SECUREAPPROTECT)* stops unauthorized access to the secure domain.

Note: If the device has activated both APPROTECT and ERASEPROTECT, it cannot be recovered without a proper software solution. See [Checking protection status](#) on page 13 for more information.

APPROTECT

APPROTECT blocks debugger read/write access to all CPU registers and memory mapped addresses.

To check if APPROTECT is already enabled, read the UICR.APPROTECT(0x00FF8000) register. If this register has a value other than 0x50FA50FA, it is protected. If the register shows the device as unprotected, write any value other than 0x50FA50FA to it. The protection activates after a reset. If you are activating ERASEPROTECT or SECUREAPPROTECT, wait to reset until all the protections are set.

ERASEPROTECT

ERASEPROTECT blocks NVMC ERASEALL and CTRL-AP.ERASEALL functionality.

To check if ERASEPROTECT is already enabled, read the UICR.ERASEPROTECT(0x00FF8020) register. If this register has a value other than 0xFFFFFFFF, it is protected. If the register shows the device as unprotected, write any value other than 0xFFFFFFFF to it. The protection activates after a reset. If you are activating APPROTECT or SECUREAPPROTECT, wait to reset until all the protections are set.

SECUREAPPROTECT

SECUREAPPROTECT blocks debugger read/write access to all secure CPU registers and secure memory mapped addresses.

To check if SECUREAPPROTECT is already enabled, read the UICR.SECUREAPPROTECT(0x00FF801C) register. If this register has a value other than 0x50FA50FA, it is protected. If the register shows the device as unprotected, write any value other than 0x50FA50FA to it. The protection activates after a reset. If you are activating APPROTECT or ERASEPROTECT, wait to reset until all the protections are set.

8 Disconnecting

Use the standard *SWD* Arm CoreSight *DAP* protocol to exit the debug interface mode.

This is managed using the built-in *CxxxPWRUPREQ* and *CxxxPWRUPACK* features found in the Arm CoreSight *DAP*. When the debugger stops requesting the debug domain or the complete system to be powered up, the device exits the debug interface mode.

We recommend a reset of the device that results in cold boot after programming by doing a pin reset or a power cycle.

9 Troubleshooting

The nRF5340 devices can be reprogrammed by erasing pages. Reprogramming is also dependent on the current nRF5340 security setting.

9.1 Checking protection status

In the nRF5340 devices, access port protection is enabled by default, but the devices can also have secure access port protection and erase protection.

For more information on access port protection, see [Enabling device protection](#) on page 11.

To check if your device is protected, read the following registers in the AHB-AP:

- AHB-AP Control/Status Word (CSW) register – use this register to read the *APPROTECT* status. This register is defined in the [Arm® CoreSight SoC-400 Technical Reference Manual](#). Use the following fields to check the access port protection status:
 - DgbStatus field (bit 6 in AHB-AP.CSW) – indicates if AHB transfers are permitted. If the value of AHB-AP.CSW->DgbStatus is 1, then AHB transfers are allowed, and the device does not have APPROTECT.
 - SPIStatus field (bit 23 in AHB-AP.CSW) – indicates if secure protection is enabled. If the value of AHB-AP.CSW->SPIStatus is 1, *SECUREAPPROTECT* is not set and both secure and non-secure transfers are allowed.
- CTRL-AP.ERASEPROTECT.STATUS – If access port protection is not enabled, use this register to check if *ERASEPROTECT* is set. If both APPROTECT and ERASEPROTECT are set, the device cannot be unlocked unless programmed software changes the settings.

9.1.1 APPROTECT and ERASEPROTECT are enabled

If both *APPROTECT* and *ERASEPROTECT* are enabled, access port 0 and the ERASEALL functionality are unavailable.

To unlock the device, it must have compatible firmware that provides a 32-bit non-zero KEY value to ERASEPROTECT.DISABLE. When both the debugger and firmware provide the same 32-bit non-zero KEY value to ERASEPROTECT.DISABLE, the device does a *CTRL-AP* erase all operation. The access port is re-enabled on the next reset once the erase sequence is done.

9.1.2 Only APPROTECT is enabled

If *APPROTECT* is enabled on the device, access port 0 is unavailable.

The only way to reopen or unlock the device is to issue an ERASEALL command through the *CTRL-AP* access port and then issue a pin reset. This will erase the entire code flash, the *UICR* area of the device, and the entire RAM. This method of erasing is slower than performing an *NVMC* erase all since it also must erase all RAM, but if APPROTECT is enabled, it is the only way to unlock the device.

9.1.2.1 Erasing all through CTRL-AP

Use the standard *SWD* Arm CoreSight *DAP* protocol to erase all while *CTRL-AP* is still selected by the debug port.

1. Write 0x00000001 to the ERASEALL register (0x004) of CTRL-AP.
This will start the ERASEALL operation which erases all flash and RAM on the device.
2. Read the ERASEALLSTATUS register (0x008) of the CTRL-AP until the value read is 0x00 (Ready) or wait 15 seconds after the ERASEALL write has expired.

3. Issue a pin reset.

9.1.2.2 Halting the CPU

Use the standard *SWD* Arm CoreSight *DAP* protocol to issue a halt command to the chip.

An application running on the device that was previously programmed can use the *WDT*. The default configuration of the *WDT* will pause it, if the CPU is halted.

9.1.2.3 Reloading the watchdog timer

If the *WDT* is configured not to stop on a halt command, it must be reloaded periodically to prevent it from resetting the domain.

1. Read the *WDT.RUNSTATUS* register (0x50018400) to check if the *WDT* is running.
If the least significant bit is 1, the *WDT* is running.
2. Read the **HALT** field of the *WDT.CONFIG* (0x5001850C) register to check if the *WDT* halts with the CPU.
If the fourth least significant bit is 1, the *WDT* does not halt.
3. Reload the *WDT* by doing the following:
 - a) Identify an enabled reload request in *WDT.RREN* (0x50018508).
If value of the bit *n* in *WDT.RREN* is 1, the reload request *n* in *WDT.RR[n]* (0x50018600 + (*n* × 0x4)) can be used to reload the watchdog counter to the value written in *WDT.CRV* (0x50018504).
 - b) Reload the watchdog timer by writing 0x6E524635 to the *WDT.RR[n]* register.

9.1.2.4 Reading FICR

*Factory Information Configuration Registers (FICR)*s are preprogrammed in the factory and cannot be erased by the user. These registers contain chip-specific information and configuration.

Using the standard *SWD* Arm CoreSight *DAP* protocol:

1. Read the *INFO.CODEPAGESIZE* register (0x00FF0220) of the *FICR*.
The value of this register contains the code memory page size in hexadecimal format, so 0x00001000 stored in this register corresponds to a page size of 4096 bytes.
2. Read the *INFO.CODESIZE* register (0x00FF0224) of the *FICR*.
The value of this register contains the number of pages in code memory in hexadecimal format, so 0x0000100 stored in this register corresponds to 256 total pages in flash memory.

Note: Total flash memory (in bytes) = *INFO.CODEPAGESIZE* * *INFO.CODESIZE*. This information is used later to determine the valid range of addresses to program.

9.1.3 APPROTECT is disabled

The *UICRs* have not been previously configured to enable access port protection.

If the device is in secure protection, you cannot read *FICRs* or disable the *System Protection Unit (SPU)*. To disable *SECUREAPPROTECT*, see [Erasing all through CTRL-AP](#) on page 13.

In some cases, you can assume that the entire flash has already been erased. If the flash is already erased and the device has never been programmed before, go to [Programming](#) on page 10.

9.1.3.1 Reading FICR

FICRs are preprogrammed in the factory and cannot be erased by the user. These registers contain chip-specific information and configuration.

Using the standard *SWD* Arm CoreSight *DAP* protocol:

1. Read the *INFO.CODEPAGESIZE* register (0x00FF0220) of the *FICR*.

The value of this register contains the code memory page size in hexadecimal format, so 0x00001000 stored in this register corresponds to a page size of 4096 bytes.

2. Read the INFO.CODESIZE register (0x00FF0224) of the FICR.

The value of this register contains the number of pages in code memory in hexadecimal format, so 0x0000100 stored in this register corresponds to 256 total pages in flash memory.

Note: Total flash memory (in bytes) = INFO.CODEPAGESIZE * INFO.CODESIZE. This information is used later to determine the valid range of addresses to program.

9.1.3.2 Halting the CPU

Use the standard *SWD* Arm CoreSight *DAP* protocol to issue a halt command to the chip.

An application running on the device that was previously programmed can use the *WDT*. The default configuration of the *WDT* will pause it, if the CPU is halted.

9.1.3.3 Reloading the watchdog timer

If the *WDT* is configured not to stop on a halt command, it must be reloaded periodically to prevent it from resetting the domain.

1. Read the WDT.RUNSTATUS register (0x50018400) to check if the *WDT* is running.
If the least significant bit is 1, the *WDT* is running.
2. Read the **HALT** field of the WDT.CONFIG (0x5001850C) register to check if the *WDT* halts with the CPU.
If the fourth least significant bit is 1, the *WDT* does not halt.
3. Reload the *WDT* by doing the following:
 - a) Identify an enabled reload request in WDT.RREN (0x50018508).
If value of the bit *n* in WDT.RREN is 1, the reload request *n* in WDT.RR[n] (0x50018600 + (n × 0x4)) can be used to reload the watchdog counter to the value written in WDT.CRV (0x50018504).
 - b) Reload the watchdog timer by writing 0x6E524635 to the WDT.RR[n] register.

9.1.3.4 Disabling SPU

The *SPU* protects memories from illegal erases, writes, and accesses.

An attempt to erase, write, or access a protected memory region leads to a secure fault. System protection can be turned off in the debug mode by configuring the relevant FLASHREGION[n].PERM for flash and RAMREGION[n].PERM for RAM.

To disable the *SPU*, the processor must be reset and stopped in the reset vector using the Flash Patch and Breakpoint unit (FPB). This disables any *SPU* protection and prevents any firmware from re-enabling the *SPU*.

See [nRF5340 SPU – System protection unit](#) for more information.

9.1.3.5 Check flash region security attribute

For every flash page, there is a security attribute that controls if that region is secure or non-secure.

Check the corresponding SPU.FLASHREGION[n].PERM (0x5003600 + (n*4)), where *n* is the flash page. If the 4th bit is 0, the flash region is in non-secure, otherwise it is secure.

9.2 Erasing

The flash can be erased by either erasing page by page or erasing all pages.

An erase all operation takes the same amount of time as erasing two pages one by one. With more than 2 flash pages, erasing all is more efficient than erasing page by page. If a region of the chip has

been preprogrammed, you can erase the flash you intend to program page by page and then write those addresses with data leaving pre-programmed flash untouched. If the value of all flash addresses is 0xFFFFFFFF, skip this procedure.

9.2.1 Erasing all

Use the standard *SWD* Arm CoreSight *DAP* protocol to erase all pages.

1. If *SECUREAPPROTECT* is enabled, erase all is disabled. To erase, see either [Erasing page by page](#) on page 16 or [Erasing all through CTRL-AP](#) on page 13 to disable *SECUREAPPROTECT*.
2. Set the CONFIG register of the NVMC to *WEN.Een* by writing 0x00000002 to the following addresses:
 - Application core - 0x50039504
 - Network core - 41080504

This configures the non-volatile memory for erasing.

3. Read the READY register of the NVMC until the value is 0x00000001. Read the following addresses:
 - Application core - 0x50039400
 - Network core - 0x41080400

When this value is read, the NVMC is ready and not currently performing any operations.

4. Set ERASEALL register of the NVMC to *ERASEALL.Erase* by writing 0x00000001 to the following addresses:
 - Application core - 0x5003950C
 - Network core - 4108050C

This erases all non-volatile memory including *UICR* registers.

5. Read the READY register of the NVMC until the value is 0x00000001 before continuing to ensure the erase all operation has completed. Read the following addresses:
 - Application core - 0x50039400
 - Network core - 0x41080400
6. Set the CONFIG register of the NVMC to *WEN.Ren* by writing 0x00000000 to following addresses:
 - Application core - 0x50039504
 - Network core - 41080504

This configures the non-volatile memory back to read-only.

9.2.2 Erasing page by page

Use the standard *SWD* Arm CoreSight *DAP* protocol to erase page by page.

9.2.2.1 SECUREAPPROTECT disabled on network core

Use this procedure to erase page by page if *SECUREAPPROTECT* is disabled on the network core.

Before you begin, check *SPU* if the flash page is secure or non-secure (see [Check flash region security attribute](#) on page 15).

1. Set the CONFIG register of the NVMC to *WEN.Een* by writing 0x00000002 to the following addresses:
 - Application core - 0x50039504
 - Network core - 41080504

This configures the non-volatile memory for erasing.

2. Read the READY register of the NVMC until the value is 0x00000001. Read the following addresses:

- Application core - 0x50039400
- Network core - 0x41080400

When this value is read, the NVMC is ready and not currently performing any operations.

3. Write 0xFFFFFFFF to the first 32-bit word in the flash page you want to be erased.
4. Read the READY register of the NVMC until the value is 0x00000001. Read the following addresses:
 - Application core - 0x50039400
 - Network core - 0x41080400
5. Repeat steps 3 and 4 until all wanted pages are erased.
6. Set the CONFIG register of the NVMC to WEN . Ren by writing 0x00000000 to the following addresses:
 - Application core - 0x50039504
 - Network core - 41080504

This configures the non-volatile memory back to read-only.

Note: The *UICR* flash page cannot be erased by using erase page. It can be erased only by an erase all operation.

9.2.2.2 SECUREAPPROTECT enabled

Use this procedure to erase page by page if *SECUREAPPROTECT* is enabled.

Note: This procedure is only valid for the application core.

1. Write 0x00000002 to the CONFIGNS register (0x40039584) of the NVMC.
This configures the non-volatile memory for erasing.
2. Read the READY register (0x40039400) of the NVMC until the value is 0x00000001.
When this value is read, the NVMC is ready and not currently performing any operations.
3. Write 0xFFFFFFFF to the first 32-bit word in the flash page you want to be erased.
4. Read the READY register (0x40039400) of the NVMC until the value is 0x00000001.
5. Repeat steps 3 and 4 until all wanted pages are erased.
6. Write 0x00000000 to the CONFIGNS register (0x40039584) of the NVMC.
This configures the non-volatile memory back to read-only.

Glossary

Access Port Protection (APPROTECT)

A register used to prevent read and write access to all CPU registers and memory-mapped addresses.

Control Access Port (CTRL-AP)

A custom access port that enables control of the device even if other access ports in the debug access port are disabled by the access port protection.

Debug Access Port (DAP)

Provides multiple master driving ports, all accessible and controlled through a single external interface port to provide system-wide debug.

Erase Protection (ERASEPROTECT)

A register used to block NVMC ERASEALL, RRAMC ERASEALL, and CTRL-AP.ERASEALL functionality.

Factory Information Configuration Registers (FICR)

Pre-programmed registers that contain chip-specific information and configuration. FICRs cannot be erased by users.

Non-volatile Memory Controller (NVMC)

A controller used for writing and erasing the internal flash memory and the *UICR*.

Resistive Random-Access Memory Controller (RRAMC)

A controller used for writing and erasing the non-volatile memory and the *UICR*.

Secure Access Port Protection (SECUREAPPROTECT)

A register used to prevent read and write access to all secure CPU registers and secure memory-mapped addresses.

Serial Wire Debug (SWD)

A standard two-wire interface for programming and debugging Arm CPUs.

Serial Wire Debug Port (SW-DP)

An interface that provides a low pin count bi-directional connection to the DAP with a reference clock signal for synchronous operation.

System on Chip (SoC)

A microchip that integrates all the necessary electronic circuits and components of a computer or other electronic systems on a single integrated circuit.

System Protection Unit (SPU)

The central point in the system that controls access to memories, peripherals, and other resources.

User Information Configuration Registers (UICR)

Non-volatile memory registers used to configure user-specific settings.

Watchdog timer (WDT)

A timer that causes a system reset if it is not poked periodically.

Recommended reading

In addition to the information in this document, you may need to consult other documents.

Nordic documentation

The following sections in the [nRF5340 Product Specification](#) contain relevant information for programming *System on Chip (SoC)*s:

- [Memory](#) – nRF5340 devices use flash-based non-volatile memory in the code flash, *UICR*, and *FICR* memory regions. This section includes information about flash page size and number.
- [NVMC — Non-volatile memory controller](#) – has detailed specifications about timing for write/erase operations.
- [Debug and trace](#) – provides access to the on-chip debug functionality. This is a standard two-pin *SWD* interface as defined by Arm.

Other documentation

See the [CoreSight Components Technical Reference Manual](#) for more information on general concepts such as Arm CoreSight or *SWD*.

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